

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2020 P1H Worked Solutions

Jan 2020 | P1H

25 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

- 1** The point A has coordinates $(5, -4)$
 The point B has coordinates $(13, 1)$

(a) Work out the coordinates of the midpoint of AB .

(.....,)
 (2)

Line L has equation $y = 2 - 3x$

(b) Write down the gradient of line L .

.....
 (1)

Line L has equation $y = 2 - 3x$

- (c) Does the point with coordinates $(100, -302)$ lie on line L ?
 You must give a reason for your answer.

.....
 (1)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION**Q#: 1****Marks: 4****TOPIC: Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the midpoint**(a) The midpoint of $A(5, -4)$ and $B(13, 1)$ is

$$\left(\frac{5 + 13}{2}, \frac{-4 + 1}{2} \right)$$

$$= (9, -1.5)$$

2. (b)

(b)

$$y = 2 - 3x$$

3. The gradient isThe gradient is -3 .**4. Calculate value**(c) For $x = 100$,

$$y = 2 - 3(100) = -298$$

5. Calculate valueThe point has $y = -302$, not -298 , so it does not lie on the line.**FINAL ANSWER** $(9, -1.5)$, gradient -3 , and no.

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

Jan 2020 P1H - Jan 2020 | P1H

- 2** Find the lowest common multiple (LCM) of 28 and 105

(Total for Question 2 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 2** **Marks: 2****TOPIC: Prime Factors, HCF & LCM**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Write both numbers as products**

Write both numbers as products of prime factors:

$$28 = 2^2 \times 7$$

$$105 = 3 \times 5 \times 7$$

2. Find the LCM

For the lowest common multiple, take the highest power of each prime:

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

FINAL ANSWER

420

PAST PAPER QUESTION**Q#: 3****Marks: 4****TOPIC: Area & Perimeter**

Jan 2020 P1H - Jan 2020 | P1H

- 3 The diagram shows a shape.

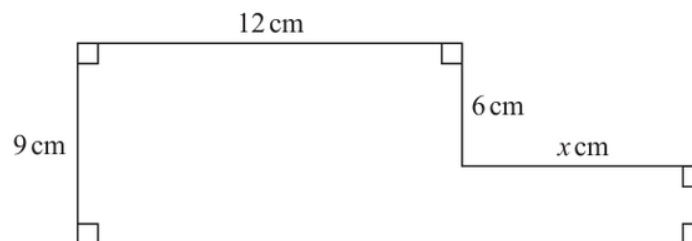


Diagram **NOT**
accurately drawn

The shape has area 129 cm^2

Work out the value of x .

$x =$

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 4****TOPIC: Area & Perimeter**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Split the shape into two**

Split the shape into two rectangles.

2. The left rectangle has area

The left rectangle has area

$$12 \times 9 = 108 \text{ cm}^2$$

3. The right rectangle has height

The right rectangle has height

$$9 - 6 = 3 \text{ cm}$$

4. Calculate area

So its area is

$$3x$$

5. Total area

Total area:

$$108 + 3x = 129$$

$$3x = 21$$

$$x = 7$$

FINAL ANSWER

7 cm.

PAST PAPER QUESTION**Q#: 4** **Marks: 7****TOPIC: Statistics Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

- 4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean weight of the 40 babies.

..... kg
(4)

One of the 40 babies is going to be chosen at random.

- (c) Find the probability that this baby has a weight of more than 5 kg.

.....
(2)

(Total for Question 4 is 7 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 7****TOPIC: Statistics Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a) The greatest frequency is**

(a) The greatest frequency is 16, so the modal class is

$$3 < w \leq 4$$

2. (b) Use class midpoints

(b) Use class midpoints:

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

3. (c) Weights greater than kg

(c) Weights greater than 4 kg:

$$9 + 2 + 1 = 12$$

$$P(w > 4) = \frac{12}{40} = \frac{3}{10}$$

FINAL ANSWER(a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

PAST PAPER QUESTION**Q#: 5****Marks: 6****TOPIC: Ratio Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

- 5** 120 children go on an activity holiday.
The ratio of the number of girls to the number of boys is 3 : 5
- On Sunday, all the children either go sailing or go climbing.
- $\frac{16}{25}$ of the boys go climbing.
- Twice as many girls go sailing as go climbing.
- Work out how many children go sailing on Sunday.

(Total for Question 5 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 5****Marks: 6**TOPIC: **Ratio Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Split the ratio**

Girls : boys = 3 : 5, so there are 8 parts.

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

$$\text{boys} = 120 \times \frac{5}{8} = 75$$

2. Split the ratio

Boys climbing:

$$\frac{16}{25} \times 75 = 48$$

3. Split the ratio

Boys sailing:

$$75 - 48 = 27$$

4. Split the ratio

For the girls, twice as many go sailing as climbing, so the girls are split in the ratio

$$\text{sailing : climbing} = 2 : 1$$

5. Split the ratio

Girls sailing:

$$\frac{2}{3} \times 45 = 30$$

6. Total sailing

Total sailing:

$$27 + 30 = 57$$

FINAL ANSWER

57

PAST PAPER QUESTION**Q#: 6****Marks: 3****TOPIC: Powers, Roots & Standard Form**

Jan 2020 P1H - Jan 2020 | P1H

- 6 (a) Write 7.8×10^{-4} as an ordinary number.

.....
(1)

- (b) Work out $\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$

Give your answer in standard form.

.....
(2)

(Total for Question 6 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 6** **Marks: 3****TOPIC: Powers, Roots & Standard Form**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

$$7.8 \times 10^{-4} = 0.00078$$

2. Convert standard form

For part (b),

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

3. Convert standard form

Then

$$\begin{aligned} \frac{63000}{2.8 \times 10^{-3}} &= \frac{63000}{0.0028} \\ &= 22500000 = 2.25 \times 10^7 \end{aligned}$$

FINAL ANSWER(a) 0.00078, (b) 2.25×10^7

PAST PAPER QUESTION**Q#: 7****Marks: 9****TOPIC: Solving Linear Equations**

Jan 2020 P1H - Jan 2020 | P1H

7 (a) Expand and simplify $(m - 8)(m + 5)$
(2)(b) Factorise fully $5y + 20y^2$
(2)(c) Simplify $(p^2 + 3)^0$
(1)(d) Solve $3(2x - 5) = \frac{9 - x}{2}$

Show clear algebraic working.

 $x =$
(4)**(Total for Question 7 is 9 marks)**

PAST PAPER SOLUTION**Q#: 7****Marks: 9****TOPIC: Solving Linear Equations**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

$$(a) (m - 8)(m + 5) = m^2 - 3m - 40.$$

2. (b)

$$(b) 5y + 20y^2 = 5y(1 + 4y).$$

3. (c)

$$(c) (p^2 + 3)^0 = 1.$$

4. (d)

(d)

$$3(2x - 5) = \frac{9 - x}{2}$$

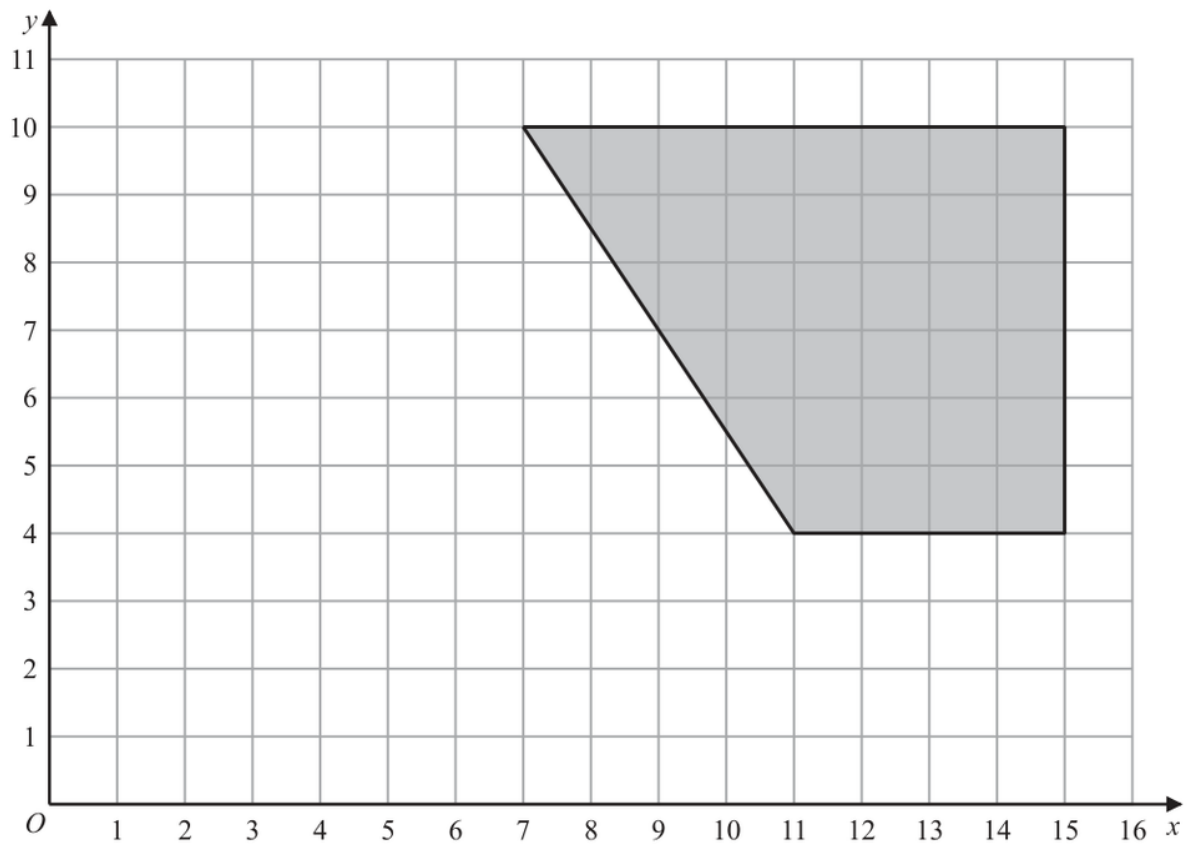
$$12x - 30 = 9 - x$$

$$13x = 39$$

FINAL ANSWER(a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

PAST PAPER QUESTION**Q#: 8****Marks: 2****TOPIC: Transformations**

Jan 2020 P1H - Jan 2020 | P1H

8

On the grid, enlarge the shaded shape with scale factor $\frac{1}{2}$ and centre (1, 2)

(Total for Question 8 is 2 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 2****TOPIC: Transformations**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use**

Use

$$\text{new point} = (1, 2) + \frac{1}{2}(\text{old point} - (1, 2))$$

2. The vertices

The vertices

$$(7, 10), (15, 10), (15, 4), (11, 4)$$

3. become

become

$$(4, 6), (8, 6), (8, 3), (6, 3)$$

FINAL ANSWERdraw the enlarged shape with vertices $(4, 6), (8, 6), (8, 3), (6, 3)$.

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

9 Here is a right-angled triangle.

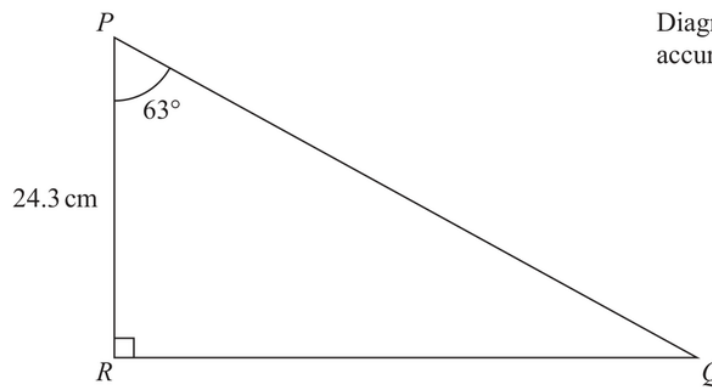


Diagram **NOT**
accurately drawn

Calculate the length of PQ .
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION**Q#: 9** **Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

$$\cos 63^\circ = \frac{24.3}{PQ}$$

$$PQ = \frac{24.3}{\cos 63^\circ} = 53.525 \dots$$

FINAL ANSWER

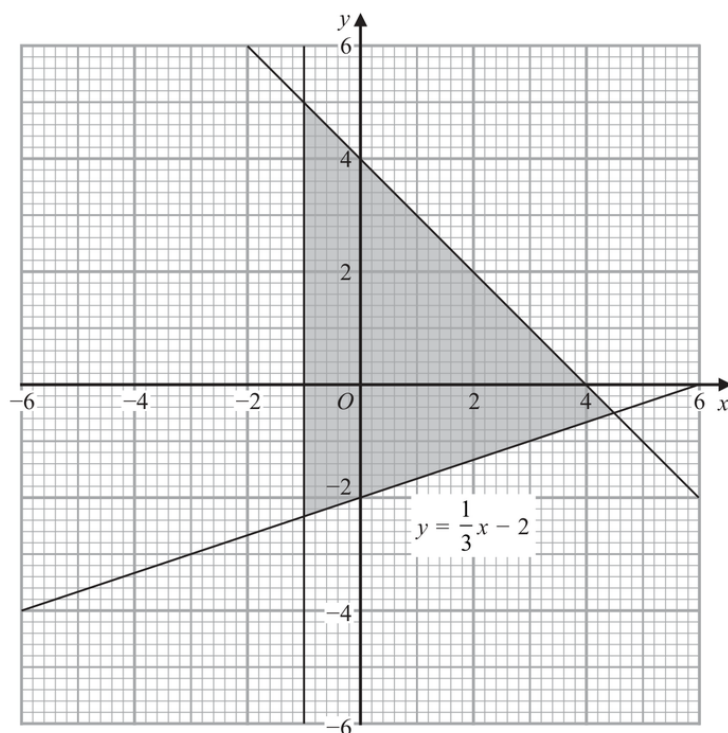
53.5 cm.

EA

PAST PAPER QUESTION**Q#: 10** **Marks: 3****TOPIC: Graphing Inequalities**

Jan 2020 P1H - Jan 2020 | P1H

- 10** The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down the three inequalities that define the shaded region.

.....

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 3****TOPIC: Graphing Inequalities**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The shaded region is left**The shaded region is left of the y -axis:

$$x \leq 0$$

2. It is below the lineIt is below the line through $(0, 4)$ and $(4, 0)$:

$$y \leq -x + 4$$

3. It is above the line

It is above the line

$$y = \frac{1}{3}x - 2$$

4. Evaluate fraction

So

$$y \geq \frac{1}{3}x - 2$$

FINAL ANSWER

$$x \leq 0, y \leq -x + 4, y \geq \frac{1}{3}x - 2.$$

PAST PAPER QUESTION**Q#: 11****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2020 P1H - Jan 2020 | P1H

- 11** Max invests \$6000 in a savings account for 3 years.
The account pays compound interest at a rate of 1.5% per year for the first 2 years.
- The compound interest rate changes for the third year.
At the end of 3 years, there is a total of \$6311.16 in the account.
- Work out the compound interest rate for the third year.
Give your answer correct to 1 decimal place.

..... %

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 3****TOPIC:** Compound Interest & Depreciation

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use compound interest**

After the first 2 years,

$$6000(1.015)^2 = 6181.35$$

2. Let the third year interestLet the third-year interest rate be $r\%$.

$$6181.35 \left(1 + \frac{r}{100} \right) = 6311.16$$

$$1 + \frac{r}{100} = \frac{6311.16}{6181.35} = 1.021000 \dots$$

$$r = 2.1000 \dots$$

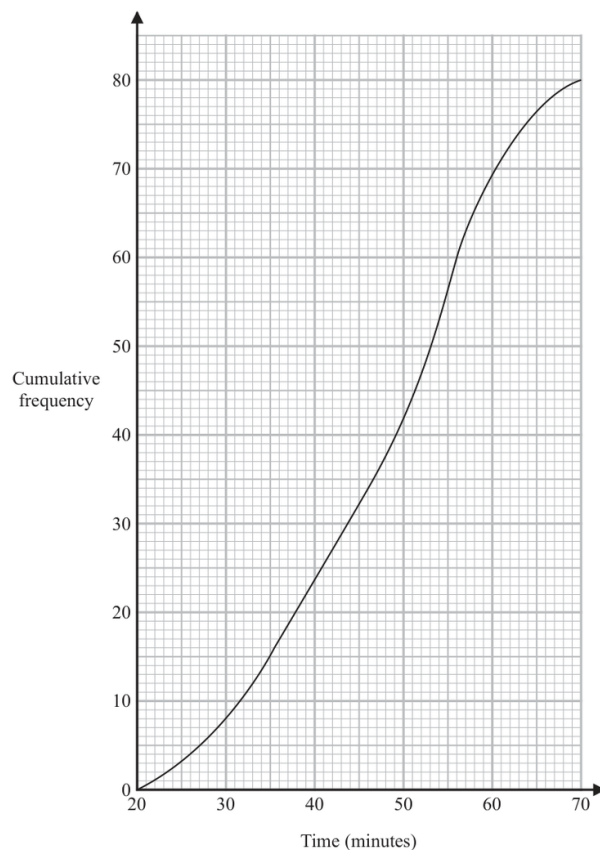
FINAL ANSWER

2.1%

PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Jan 2020 P1H - Jan 2020 | P1H

- 12 A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.



- (a) Use the graph to find an estimate for the interquartile range.

..... minutes
(2)

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

- (b) Work out an estimate for the number of men who took part in the race.

.....
(3)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. There are people**

There are 80 people.

2. Use cumulative frequency

For the interquartile range:

 Q_1 is at cumulative frequency 20 Q_3 is at cumulative frequency 60**3. Use cumulative frequency**

From the graph,

$$Q_1 \approx 38, \quad Q_3 \approx 56$$

$$\text{IQR} \approx 56 - 38 = 18$$

4. Use cumulative frequency

At 50 minutes, the cumulative frequency is about 42.

5. Use cumulative frequency

No women took 50 minutes or less, so these 42 people are men.

$$0.6 \times \text{number of men} = 42$$

$$\text{number of men} = \frac{42}{0.6} = 70$$

FINAL ANSWER

IQR about 18 minutes, and about 70 men.

PAST PAPER QUESTION**Q#: 13****Marks: 4****TOPIC: Standard & Compound Units**

Jan 2020 P1H - Jan 2020 | P1H

13 The diagram shows a solid cube.

The cube is placed on a table so that the whole of one face of the cube is in contact with the table.

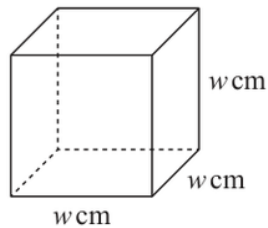


Diagram **NOT**
accurately drawn

The cube exerts a force of 56 newtons on the table.

The pressure on the table due to the cube is 0.14 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cube.

..... cm³

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 4****TOPIC: Standard & Compound Units**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate area**

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$0.14 = \frac{56}{\text{area}}$$

$$\text{area} = \frac{56}{0.14} = 400 \text{ cm}^2$$

2. The cube rests on one

The cube rests on one square face, so

$$w^2 = 400$$

$$w = 20$$

3. Volume of the cube is

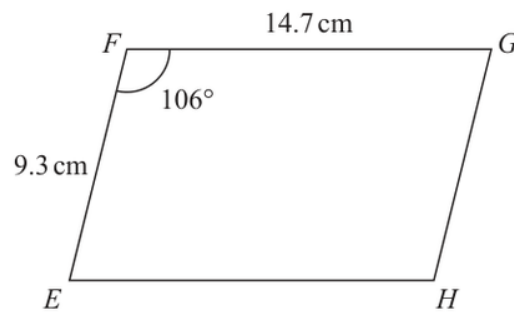
Volume of the cube is

$$20^3 = 8000$$

FINAL ANSWER8000 cm³.

PAST PAPER QUESTION**Q#: 14****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1H - Jan 2020 | P1H

14 The diagram shows parallelogram $EFGH$.Diagram **NOT**
accurately drawn

$EF = 9.3 \text{ cm}$
 $FG = 14.7 \text{ cm}$
 $\text{Angle } EFG = 106^\circ$

- (a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm^2
(2)

- (b) Work out the length of the diagonal EG of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm
(3)

(Total for Question 14 is 5 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

2. Answer**FINAL ANSWER**131 cm².**3. Use trigonometry**(b) In triangle EFG ,

$$EG^2 = 9.3^2 + 14.7^2 - 2(9.3)(14.7) \cos 106^\circ$$

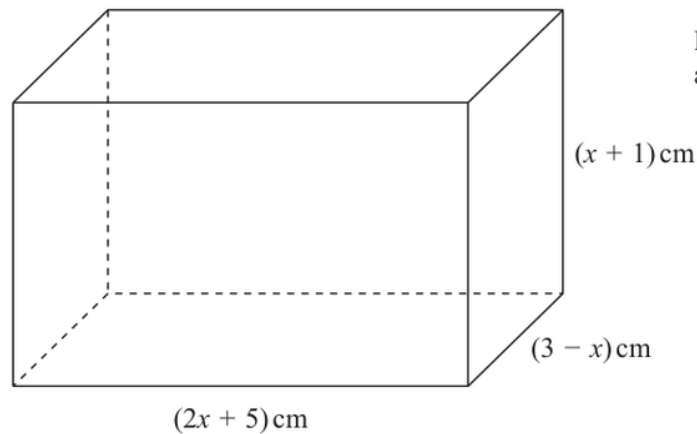
$$EG = 19.440 \dots$$

FINAL ANSWER

19.4 cm.

PAST PAPER QUESTION**Q#: 15****Marks: 8****TOPIC: Differentiation**

Jan 2020 P1H - Jan 2020 | P1H

15Diagram **NOT**
accurately drawnThe diagram shows a cuboid of volume $V\text{cm}^3$ (a) Show that $V = 15 + 16x - x^2 - 2x^3$

(3)

There is a value of x for which the volume of the cuboid is a maximum.

- (b) Find this value of x .
 Show your working clearly.
 Give your answer correct to 3 significant figures.

 $x = \dots\dots\dots$
 (5)
(Total for Question 15 is 8 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 8****TOPIC: Differentiation**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The volume is**

The volume is

$$V = (2x + 5)(x + 1)(3 - x)$$

2. Expand brackets

First expand:

$$(2x + 5)(x + 1) = 2x^2 + 7x + 5$$

$$V = (2x^2 + 7x + 5)(3 - x)$$

$$V = 15 + 16x - x^2 - 2x^3$$

3. Calculate volume

For the maximum volume,

$$\frac{dV}{dx} = 0$$

$$\frac{dV}{dx} = 16 - 2x - 6x^2$$

$$16 - 2x - 6x^2 = 0$$

$$3x^2 + x - 8 = 0$$

4. Solve quadratic equation

Using the quadratic formula,

$$x = \frac{-1 \pm \sqrt{97}}{6}$$

5. The valid positive value is

The valid positive value is

$$x = 1.4748 \dots$$

FINAL ANSWER $x = 1.47$ to 3 significant figures.

PAST PAPER QUESTION**Q#: 16****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2020 P1H - Jan 2020 | P1H

16 $P = \frac{2a - c}{d}$

$a = 58.4$ correct to 3 significant figures.

$c = 20$ correct to 2 significant figures.

$d = 3.6$ correct to 2 significant figures.

Work out the upper bound for the value of P .

Show your working clearly.

Give your answer correct to 2 decimal places.

.....
(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION**Q#: 16** **Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Estimate the value**

$$P = \frac{2a - c}{d}$$

2. Find upper bound

For the upper bound of P , make the numerator as large as possible and the denominator as small as possible.

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

$$P_{\text{upper}} = \frac{2(58.45) - 19.5}{3.55} = 27.4366 \dots$$

FINAL ANSWER

27.44

PAST PAPER QUESTION**Q#: 17****Marks: 6****TOPIC: Algebraic Roots & Indices**

Jan 2020 P1H - Jan 2020 | P1H

17 (a) Show that $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(3)

(b) Simplify fully $\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$
(3)

(Total for Question 17 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 6****TOPIC: Algebraic Roots & Indices**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

First,

$$\sqrt{12} = 2\sqrt{3}$$

2. Simplify surd

So

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

3. Simplify surd

Now square:

$$\begin{aligned}(6 + 4\sqrt{3})^2 &= 36 + 48\sqrt{3} + 48 \\ &= 84 + 48\sqrt{3}\end{aligned}$$

4. Factor out

Factor out 12:

$$84 + 48\sqrt{3} = 12(7 + 4\sqrt{3})$$

5. Simplify surd

For part (b),

$$\begin{aligned}\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}} &= \left(\frac{t^{15}}{27a^{12}}\right)^{\frac{2}{3}} \\ &= \frac{t^{10}}{9a^8}\end{aligned}$$

FINAL ANSWER(a) shown, (b) $\frac{t^{10}}{9a^8}$

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Probability Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

18 There are 16 sweets in a bowl.

4 of the sweets are blackcurrant.

5 of the sweets are lemon.

7 of the sweets are orange.

Anna, Ravi and Sam each take at random one sweet from the bowl.

Work out the probability that the 5 lemon sweets are still in the bowl.

(Total for Question 18 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: Probability Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. There are sweets and are**

There are 16 sweets and 5 are lemon, so 11 are not lemon.

$$\begin{aligned}P(\text{no lemon}) &= \frac{11}{16} \times \frac{10}{15} \times \frac{9}{14} \\&= \frac{33}{112}\end{aligned}$$

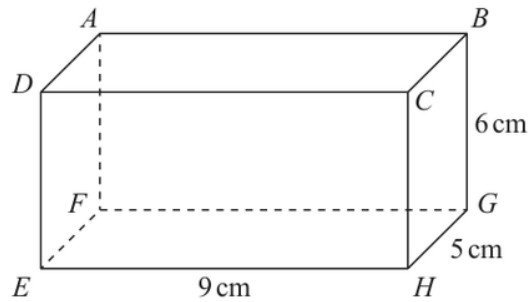
FINAL ANSWER

$$\frac{33}{112}.$$

EA

PAST PAPER QUESTION**Q#: 19****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

19 The diagram shows a cuboid $ABCDEFGH$.Diagram **NOT**
accurately drawn $EH = 9\text{ cm}$, $HG = 5\text{ cm}$ and $GB = 6\text{ cm}$.Work out the size of the angle between AH and the plane $EFGH$.
Give your answer correct to 3 significant figures.

o

(Total for Question 19 is 4 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The projection of onto the

The projection of AH onto the base plane is the diagonal of the base rectangle:

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

2. Use trigonometry

The vertical height is 6 cm, so

$$\tan \theta = \frac{6}{\sqrt{106}}$$

$$\theta = 30.232 \dots^\circ$$

FINAL ANSWER

30.2°.

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Graphs of Functions**

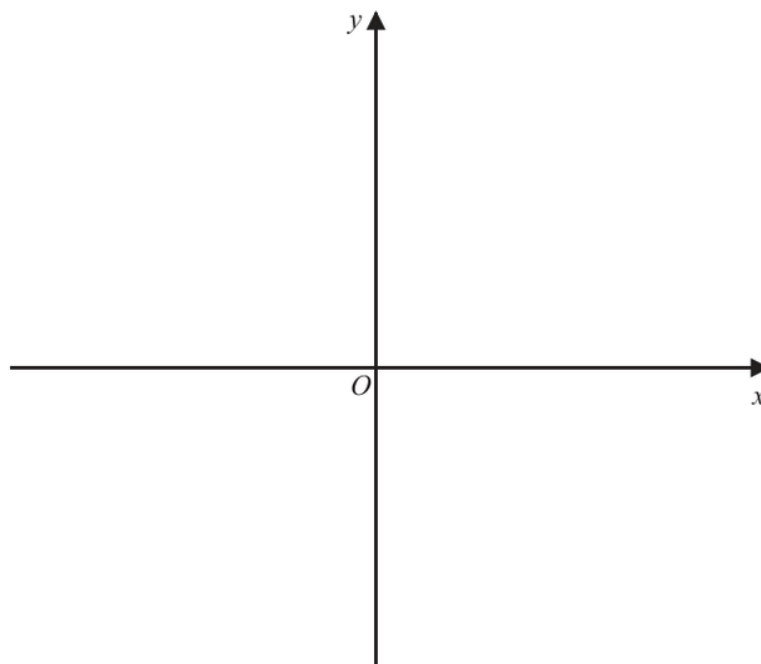
Jan 2020 P1H - Jan 2020 | P1H

20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.

On your sketch show clearly, in terms of a ,

- (i) the coordinates of any points of intersection of **C** with the coordinate axes,
- (ii) the coordinates of the turning point.



(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Graphs of Functions**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$y = 4(x - 1)^2 - a$$

2. The turning point is read

The turning point is read directly from the completed square form:

$$(1, -a)$$

3. Find intercepts

For the y -intercept, put $x = 0$:

$$y = 4(0 - 1)^2 - a$$

$$y = 4 - a$$

4. Find intercepts

So the y -intercept is

$$(0, 4 - a)$$

5. Find intercepts

For the x -intercepts, put $y = 0$:

$$4(x - 1)^2 - a = 0$$

$$4(x - 1)^2 = a$$

$$(x - 1)^2 = \frac{a}{4}$$

$$x - 1 = \pm \frac{\sqrt{a}}{2}$$

$$x = 1 \pm \frac{\sqrt{a}}{2}$$

6. Find intercepts

So the x -intercepts are

$$\left(1 - \frac{\sqrt{a}}{2}, 0\right) \quad \text{and} \quad \left(1 + \frac{\sqrt{a}}{2}, 0\right)$$

7. Solve inequality

Since $a > 4$, the turning point is below the x -axis and the curve crosses the x -axis twice.

FINAL ANSWER

turning point $(1, -a)$, y -intercept $(0, 4 - a)$, x -intercepts $(1 - \frac{\sqrt{a}}{2}, 0)$ and $(1 + \frac{\sqrt{a}}{2}, 0)$

EA

PAST PAPER QUESTION

Q#: 20

Marks: 4

TOPIC: **Graphs of Functions**

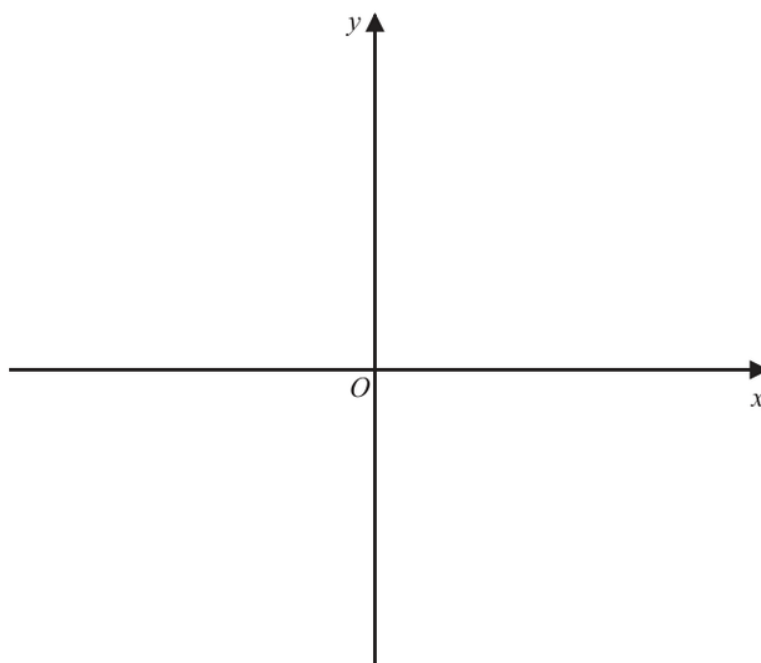
Jan 2020 P1H - Jan 2020 | P1H

20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.

On your sketch show clearly, in terms of a ,

- (i) the coordinates of any points of intersection of **C** with the coordinate axes,
- (ii) the coordinates of the turning point.



(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Graphs of Functions**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$y = 4(x - 1)^2 - a$$

2. The turning point is read

The turning point is read directly from the completed square form:

$$(1, -a)$$

3. Find intercepts

For the y -intercept, put $x = 0$:

$$y = 4(0 - 1)^2 - a$$

$$y = 4 - a$$

4. Find intercepts

So the y -intercept is

$$(0, 4 - a)$$

5. Find intercepts

For the x -intercepts, put $y = 0$:

$$4(x - 1)^2 - a = 0$$

$$4(x - 1)^2 = a$$

$$(x - 1)^2 = \frac{a}{4}$$

$$x - 1 = \pm \frac{\sqrt{a}}{2}$$

$$x = 1 \pm \frac{\sqrt{a}}{2}$$

6. Find intercepts

So the x -intercepts are

$$\left(1 - \frac{\sqrt{a}}{2}, 0\right) \quad \text{and} \quad \left(1 + \frac{\sqrt{a}}{2}, 0\right)$$

7. Solve inequality

Since $a > 4$, the turning point is below the x -axis and the curve crosses the x -axis twice.

FINAL ANSWER

turning point $(1, -a)$, y -intercept $(0, 4 - a)$, x -intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Functions**

Jan 2020 P1H - Jan 2020 | P1H

21 The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geq -2$

Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Functions**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$f(x) = x^2 - 2x$$

$$g(x) = x + 3$$

$$h(x) = fg(x) = f(g(x))$$

$$h(x) = (x + 3)^2 - 2(x + 3)$$

$$h(x) = x^2 + 4x + 3$$

$$h(x) = (x + 2)^2 - 1$$

2. Let

Let

$$y = (x + 2)^2 - 1$$

3. Find inverse functionSince the domain is $x \geq -2$, take the positive square root:

$$x + 2 = \sqrt{y + 1}$$

$$x = \sqrt{y + 1} - 2$$

FINAL ANSWER

$$h^{-1}(x) = \sqrt{x + 1} - 2$$

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Functions**

Jan 2020 P1H - Jan 2020 | P1H

21 The functions f and g are such that

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Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Functions**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$f(x) = x^2 - 2x$$

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FINAL ANSWER

$$h^{-1}(x) = \sqrt{x + 1} - 2$$

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$

J is the point with coordinates $(j, 15)$ where $j < 0$

K is the point with coordinates $(6, k)$

M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the midpoint**Since M is the midpoint of JK ,

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2} \right)$$

2. Find the gradientThe gradient of HM is 2, and $H = (-4, 1)$.

$$\frac{\frac{15+k}{2} - 1}{\frac{j+6}{2} + 4} = 2$$

$$\frac{k+13}{j+14} = 2$$

$$k = 2j + 15$$

3. Simplify surd

Also,

$$JK = \sqrt{80}$$

4. Calculate value

So

$$(6-j)^2 + (k-15)^2 = 80$$

5. SubstituteSubstitute $k = 2j + 15$:

$$(6-j)^2 + (2j)^2 = 80$$

$$j^2 - 12j + 36 + 4j^2 = 80$$

$$5j^2 - 12j - 44 = 0$$

$$(5j+10)(j-4.4) = 0$$

6. Solve inequalitySince $j < 0$,

$$j = -2$$

7. Calculate value

Then

$$k = 2(-2) + 15 = 11$$

FINAL ANSWER

$$j = -2, k = 11$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

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M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

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